

SIDE VIEW

from the right · scale 0.94 : 1 · dashed = hidden · purple = mechanism

HEAD = WINDING KEY
turn it 28x clockwise
(seen from above)

NECK RATCHET
12 teeth + flexure
pawl: winds one way

RUBBER MOTOR
2 loops of #16 band
= 4 strands x 60 mm

BODY: black back
shell + white belly
cap (2 snap-fit parts)

2 M8 STEEL WASHERS
low at the back
→ CG low + behind

TAIL + Ø8 ROLLER
= 3rd wheel

HEEL WHEELS: Ø22, O-ring tyres,
0.5 mm eccentric hubs (C)

FEET hover 2 mm
(front stop)

PINION 10 T on the heel axle
(left side, under the crown)

TRIGGER PIVOT
+ crank (left
flipper, far side)

LOCK ROD in a
guide (see B)

CROWN 24 T on a
thrust face

BASE: oval tray

FRONT VIEW (armed)

scale 0.64 : 1

rocks ±1.8° (C)

HEAD TURNS
on the neck
ring

TRIGGER = LEFT
flipper; raised
= ARMED (locked)

lock rod (B)

push it
down → GO

pinion 10 T
(toy's left)

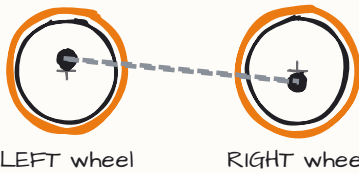
BASE: oval
tray, 42 wide

heel wheels
(track 32)

42 base

≈ 93 with the flipper raised

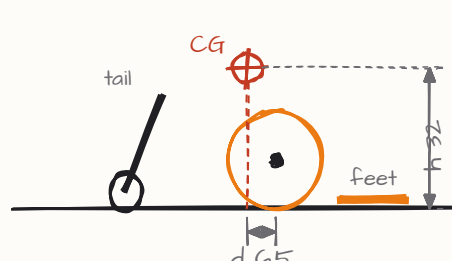
C WADDLE



LEFT wheel RIGHT wheel
axle hole 0.5 mm off-centre,
opposite sides (drawn x4)

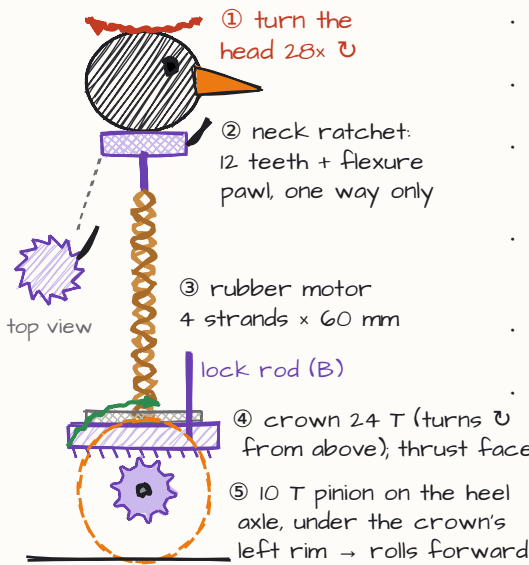
Roll = $\arctan(1 + 32 \text{ mm track}) \approx \pm 1.8^\circ$
per wheel turn; the single tail roller
lets the body rock. It follows while
 $10 \cdot \omega^2 < mg \cdot \text{track} / 2$ ($1 \approx 8.5 \times 10^{-5}$
 $\text{kg} \cdot \text{m}^2$) → below $\approx 0.55 \text{ m/s}$. Faster,
the heels take turns to hop 0.5 mm;
the slow waddle shows as it starts
and as it runs down.

D HEELS + TAIL STANCE



Stands like an emperor: on its heels
(drive wheels) and stiff tail (roller);
the feet hover 2 mm as a front stop.
Launch $a = 2.7 \text{ mN} / 44 \text{ g} \approx 0.6 \text{ m/s}^2$
just loads the tail - no wheelie.
Nose-dive needs $a = g \cdot d / h \approx 2.0 \text{ m/s}^2$;
coasting slows at $< 0.5 \text{ m/s}^2$ ✓
Side tip needs 20°; waddle $\pm 1.8^\circ$ ✓

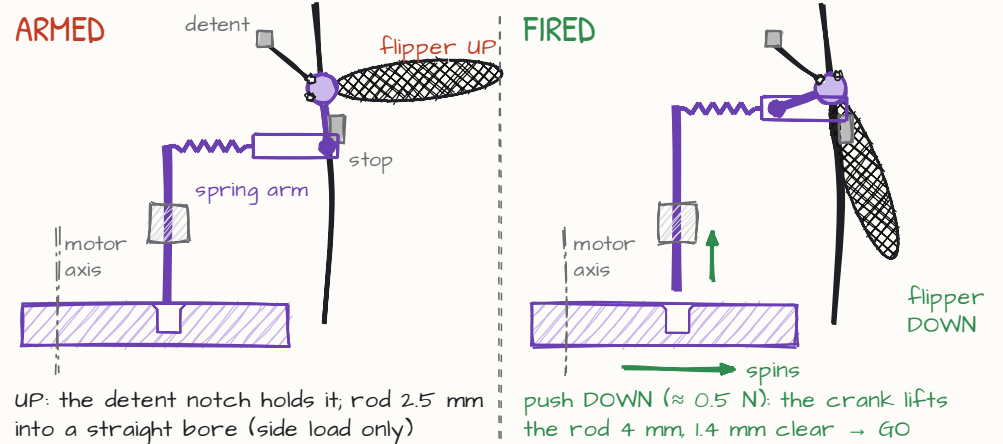
A ENERGY - TURN THE HEAD



head → ratchet → rubber → crown 24T → pinion 10T on the heel axle → wheels (x2.4)

- 1 turn the head 28x
 - 2 neck ratchet: 12 teeth + flexure pawl, one way only
 - 3 rubber motor 4 strands x 60 mm
 - 4 crown 24 T (turns 1 from above); thrust face $\approx 5.1 \text{ m}$, top 1.2 m/s; worst 3.3 m
 - 5 10 T pinion on the heel axle, under the crown's left rim → rolls forward
- Rubber: 2 loops of #16 band = 4 strands x 60 mm (0.34 g), hook to hook
 - 28 turns ($\approx 1/3$ of the ~ 90 -turn break point) → $400 \text{ J/kg} \times 0.34 \text{ g} \approx 135 \text{ mJ}$
 - 400 J/kg is our low assumption; we will measure torque vs turns (a 3rd loop adds 50 % if needed)
 - Drag: rolling 6.5 + heel-axle bores 6.2 + tail roller 4.5 = 17 mN; gears η 0.75 → 2.5 m needs 5.7 mJ (2.4x)
 - Crown 24 : pinion 10 → 2.4 wheel turns per motor turn; 28 turns = 4.6 m
 - Start force 44 mN = 2.6x drag. Time-step model (torque $\propto$ turns left): run

B TRIGGER - FLIPPER CRANK LOCK



UP: the detent notch holds it; rod 2.5 mm into a straight bore (side load only)

push DOWN ($\approx 0.5 \text{ N}$): the crank lifts the rod 4 mm, 1.4 mm clear → GO

BIOMIMICRY



Emperor penguins rest on their heels and stiff tail with toes lifted, and waddle side to side. Copied: heel drive wheels + tail roller (3-point stance), low CG, rocking wheels. Bonus: the head swivels (it is the winding key).

HOW IT PLAYS

- 1 Raise the LEFT flipper - click: the lock rod springs onto the crown.
- 2 Turn the head 28x clockwise (click-click...), stop beak-forward.
- 3 Set it down - it stands on its heels and tail, fully wound.
- 4 Push the flipper down → it waddles off, $\approx 5 \text{ m}$.

vs Concept 1: upright body, twisted rubber motor (turn key), flipper trigger.

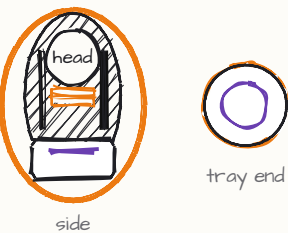
NUMBERS

- Mass $\approx 44 \text{ g}$, $\approx 86 \%$ printed PLA
- CG 32 mm high, 6.5 mm behind the heel axle (81 % on the heels)
- Motor $\approx 135 \text{ mJ}$; 2.5 m needs 5.7 mJ
- Gears ml, printed: crown 24 T → pinion 10 T (x2.4)
- Run $\approx 5.1 \text{ m}$ (worst case 3.3 m), top speed $\approx 1.2 \text{ m/s}$
- Lock-rod side load $\leq 0.17 \text{ N}$ ($1.54 \text{ mNm} \div 9 \text{ mm}$)
- Bought (9): 2 steel axles, 2 O-rings, 2 #16 bands, 2 M8 washers

RULES CHECK

- ✓ $\geq 75 \%$ printed PLA ($\approx 86 \%$)
- ✓ All parts fit the egg (box E); the built toy is bigger
- ✓ Holds its energy until the flipper is pushed down
- ✓ Trigger is a flipper crank - no removable pin, stays together
- ✓ No energy stored when apart (rubber untwisted)
- ✓ Snap / press fits: no glue, no tools
- ✓ Gears are our printed parts, not commercial toy parts

E PACKS INTO THE EGG



True scale. The 2 body parts close into a hull 66 tall x 46 x 50 holding the head, wheels, flippers, tail + roller, feet, rod, beak and bands; the oval tray (crown and washers inside) sits in the bottom of the egg. Tightest: tray 21.4 of 22.4 mm radius (right), body top 0.5 mm below the egg wall → confirm in CAD.
Assembled 109 mm tall → bigger than the egg ✓

ENGG*2100 F26 · Design & Build · Concept Sketch

Concept 2 · sheet 1 of 1

Concept 2: EMPEROR WADDLER

Theme: Biomimicry - emperor stance + waddle
Toy type: Running toy ($\geq 2.5 \text{ m}$)
Scale / units: mm · side view 0.94 : 1 on 11x17

Drawn by: _____ Signature: _____ Date: _____